

SAI JAYANTH RAJAMAHENDRAM

Software Engineer · .NET Core & TypeScript · Azure · Distributed Microservices
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SUMMARY

Backend Software Engineer with 3+ years building .NET Core microservices and distributed Azure services, with hands-on work in LLM integration and generative AI application development (OCI Generative AI Certified). Shipped production APIs with measurable latency and reliability improvements across healthcare and nonprofit domains. Comfortable owning the full delivery cycle from system design through CI/CD and live-site health.

TECHNICAL SKILLS

AI & LLM: Azure OpenAI, RAG architecture, prompt engineering, embeddings, LangChain (concepts), NLP pipelines, generative AI application development
Languages: C#, TypeScript, JavaScript, Python, SQL, Bash
Backend & Frameworks: .NET Core, ASP.NET Core, ASP.NET MVC, Entity Framework, Node.js, FastAPI
Azure: AKS, Azure DevOps, App Service, Blob Storage, Cosmos DB, Service Bus
Distributed Systems: Microservices, Redis, Kafka, REST API design, WebSockets, event-driven architecture
Containers & Orch.: Docker, Kubernetes (K8s), Helm, Service Fabric (concepts), container networking
CI/CD & DevOps: Azure DevOps Pipelines, GitHub Actions, Jenkins, live-site operations
Observability: Prometheus, Grafana, CloudWatch, ELK Stack, structured logging, alerting
Testing: xUnit, NUnit, unit testing, integration testing, TDD, continuous integration

EXPERIENCE

Software Engineer · Community Dreams Foundation · Remote	<i>Jul 2025 – Present</i>
- Integrated Azure OpenAI into the content workflow to auto-summarize donor reports, reducing manual review time by around 3 hours per week across the team. - Built 10+ .NET Core microservice endpoints serving three downstream consumers; cut average API response latency by roughly 30% by reworking query patterns and adding a lightweight in-memory cache. - Redesigned Azure DevOps CI/CD pipelines with automated integration test gates, cutting release cycle time by about 40% and removing manual handoff steps across dev, staging, and production. - Developed reusable TypeScript component libraries that standardized UI patterns across the platform and shaved roughly a quarter off new feature delivery time. - Tightened SQL Server indexing and rewrote stored procedures on core reporting queries, cutting execution time by around 35%. - Enforced unit testing standards with xUnit, pushing coverage above 85% on all new production modules and catching 15+ defects before they reached production.	
Software Engineer · Cognizant Technology Solutions · Chennai, India	<i>Jul 2021 – Aug 2023</i>
- Delivered enterprise ASP.NET Core backend services for Molina Healthcare, supporting 100K+ members in a HIPAA-compliant Azure environment with high-availability requirements. - Migrated 8 legacy SOAP services to RESTful .NET Core APIs, which noticeably cut downstream integration incidents (the team estimated around a 25% drop) and made it much easier for four internal teams to consume the services. - Rewrote 30+ T-SQL stored procedures and reworked indexing on SQL Server; the most painful report went from 12 seconds to under 3, a real win for the clinical ops team. - Integrated DocuSign REST API into the document management system, automating multi-step signature flows that previously bounced around manually across a 100K+ member base. - Worked across QA, architecture, and product through 8 consecutive Agile sprints with on-time delivery each cycle; defect escape rate dropped by roughly 20%.	
Software Engineer Intern · Cognizant Technology Solutions · Chennai, India	<i>Mar – Jul 2021</i>
- Built ASP.NET Core API modules and Angular UI components for healthcare client portals handling around 10K daily transactions. - Contributed to Azure DevOps pipeline automation across four sprint cycles, picking up end-to-end Agile delivery habits early.	

PROJECTS

LLM-Powered Document Q&A Service (RAG) .NET Core · Azure OpenAI · Cosmos DB · TypeScript · Docker · REST API
- Built a retrieval-augmented generation (RAG) pipeline in .NET Core: documents are chunked, embedded via Azure OpenAI Ada-002, and stored in Cosmos DB; at query time the top-k chunks are retrieved and injected into the GPT-4 context window, keeping end-to-end response time under 2 seconds. - Handled prompt engineering and token budget management explicitly: system prompt, retrieved context, and user query are assembled to stay within the 8K context limit, with overflow handled by chunk re-ranking rather than silent truncation. - Exposed a TypeScript REST client library so other microservices can wire in document Q&A with minimal integration effort; containerized with Docker for consistent deployment across environments.
Redis-Backed Distributed Cache Layer for Microservices .NET Core · Redis · Docker · Kubernetes · Azure
- Designed and built a shared Redis caching layer for a microservices backend in .NET Core; hot-path API latency dropped by around 40% under simulated concurrent load. - Deployed the stack on Kubernetes (AKS) with a rolling update strategy: zero downtime across three environment promotion stages.
Production Kubernetes Deployment on Azure AKS Terraform · Azure AKS · Kubernetes · Helm · Azure Load Balancer · Nginx
- Provisioned a full Azure environment from scratch using Terraform: VNet, AKS cluster, and managed node groups, so the entire stack is reproducible from a single terraform apply. - Deployed an Nginx workload on Kubernetes with a Deployment, Service, and HPA; installed Metrics Server via Helm to drive autoscaling, and exposed the app publicly through an Azure Load Balancer with zero manual portal steps.

EDUCATION

M.S. Computer Science · Illinois Institute of Technology, Chicago, IL	<i>May 2025</i>
B.E. Computer Science & Engineering · Anna University, India	<i>Apr 2021</i>

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Multicloud Architect Professional · Oracle Cloud Infrastructure 2024 Generative AI Certified Professional
Microsoft Azure Fundamentals (AZ-900) · AWS Certified Cloud Practitioner · Oracle Cloud Infrastructure Foundations (2024–2026)